Homework 09

The kernel-tuning question moved here from Homework 08. A solution for Question 1 will be available after Monday, October 26, 2026. No solution will be provided for Question 2.

Question 1: Choosing an SVM kernel

Use `saheart.csv` from the combined ZIP. As in Homework 7, predict `chd` using the same nine predictors, coding `famhist` as 1 for `Absent` and 2 for `Present`.

Compare SVMs with a linear kernel, polynomial kernels of degrees 2 (quadratic), 3, and 4, and a radial basis kernel. Choose **no more than 20 candidate settings in total** across these kernels. Vary the cost C for each kernel and the radial scale γ for the radial kernel. State any other kernel settings you hold fixed.

Divide the observations into five folds yourself and use the same folds for all candidates. Standardize predictors using only the fitting observations in each fold. Report the candidate settings and their mean cross-validation misclassification rates. Which kernel and settings do you prefer? Give a brief justification.

```
library(e1071)
set.seed(432)

d <- read.csv("data/saheart.csv")
d$famhist <- ifelse(d$famhist == "Absent", 1, 2)
d$chd <- factor(d$chd)
xvars <- setdiff(names(d), "chd")

# my own 5 folds, shared by all candidates
folds <- sample(rep(1:5, length.out = nrow(d)))

# 20 candidates; fixed: coef0 = 1, polynomial gamma = 1/p (e1071 default),
# radial uses gamma below, scale = FALSE in svm() since I standardize manually
grid <- rbind(
  data.frame(kernel = "linear",     degree = NA, gamma = NA, cost = c(0.01, 0.1, 1, 10)),
  data.frame(kernel = "polynomial", degree = 2,  gamma = NA, cost = c(0.1, 1, 10)),
  data.frame(kernel = "polynomial", degree = 3,  gamma = NA, cost = c(0.1, 1, 10)),
  data.frame(kernel = "polynomial", degree = 4,  gamma = NA, cost = c(0.1, 1, 10)),
  expand.grid(kernel = "radial", degree = NA, gamma = c(0.01, 0.05, 0.2),
              cost = c(1, 10))[, c("kernel","degree","gamma","cost")]
)
nrow(grid)  # 4 + 9 + 6 = 19
```

```
[1] 19
```

```
cv_err <- function(k, deg, gam, cost) {
  errs <- sapply(1:5, function(f) {
    tr <- d[folds != f, ]; te <- d[folds == f, ]
    mu <- colMeans(tr[xvars]); s <- apply(tr[xvars], 2, sd)  # fitting data only
    trx <- scale(tr[xvars], mu, s); tex <- scale(te[xvars], mu, s)
    args <- list(x = trx, y = tr$chd, kernel = k, cost = cost,
                 scale = FALSE, coef0 = 1)
    if (k == "polynomial") args$degree <- deg
    if (k == "radial") args$gamma <- gam
```

```
    fit <- do.call(svm, args)
    mean(predict(fit, tex) != te$chd)
  })
  mean(errs)
}

grid$cv_error <- mapply(cv_err, grid$kernel, grid$degree, grid$gamma, grid$cost)
grid[order(grid$cv_error), ]
```

```
       kernel degree gamma  cost  cv_error
14     radial     NA  0.01  1.00 0.2618513
17     radial     NA  0.01 10.00 0.2684198
5  polynomial      2    NA  0.10 0.2705937
15     radial     NA  0.05  1.00 0.2727676
8  polynomial      3    NA  0.10 0.2748948
1      linear     NA    NA  0.01 0.2769752
2      linear     NA    NA  0.10 0.2791491
6  polynomial      2    NA  1.00 0.2792660
3      linear     NA    NA  1.00 0.2813230
4      linear     NA    NA 10.00 0.2813230
11 polynomial      4    NA  0.10 0.2813698
16     radial     NA  0.20  1.00 0.2878915
7  polynomial      2    NA 10.00 0.2944367
18     radial     NA  0.05 10.00 0.3009116
10 polynomial      3    NA 10.00 0.3159888
9  polynomial      3    NA  1.00 0.3182796
12 polynomial      4    NA  1.00 0.3398784
19     radial     NA  0.20 10.00 0.3441795
13 polynomial      4    NA 10.00 0.3593502
```

```
knitr::kable(grid[order(grid$cv_error), ], row.names = FALSE, digits = 4)
```

kernel	degree	gamma	cost	cv_error
radial	NA	0.01	1.00	0.2619
radial	NA	0.01	10.00	0.2684
polynomial	2	NA	0.10	0.2706
radial	NA	0.05	1.00	0.2728
polynomial	3	NA	0.10	0.2749
linear	NA	NA	0.01	0.2770
linear	NA	NA	0.10	0.2791
polynomial	2	NA	1.00	0.2793
linear	NA	NA	1.00	0.2813
linear	NA	NA	10.00	0.2813
polynomial	4	NA	0.10	0.2814
radial	NA	0.20	1.00	0.2879
polynomial	2	NA	10.00	0.2944
radial	NA	0.05	10.00	0.3009
polynomial	3	NA	10.00	0.3160
polynomial	3	NA	1.00	0.3183
polynomial	4	NA	1.00	0.3399
radial	NA	0.20	10.00	0.3442
polynomial	4	NA	10.00	0.3594

Fixed settings: `coef0 = 1` for polynomial kernels, polynomial `gamma = 1/p` (the e1071 default), `scale = FALSE` in `svm()` because I standardized by hand within each fold, and the same 5 folds for all 19 candidates.

Preferred model: the radial kernel with $\gamma = 0.01$ and $C = 1$, which has the lowest mean CV misclassification rate (about 26.2%).

Justification: The best radial model beats the best quadratic (27.1%) and the best linear (27.7%) by less than 2 percentage points. That is only a handful of observations out of 462, so the top few models are close to tied. Still, the

pattern is clear. Error rises as flexibility rises: degree 3 and 4 polynomials, larger γ (0.2), and larger C (10) all do worse, which indicates overfitting. A small γ makes the radial boundary smooth and close to linear, so the winner is effectively a mildly nonlinear, well-regularized classifier. I choose it because it has the lowest CV error and sits in the low-flexibility region. A linear SVM would be a reasonable, simpler alternative given how small the gap is.

Question 2: Write and use a short SVM analysis skill

In Homework 8, you reviewed another student's skill. Review the comments and suggestions you received on your own Homework 7 skill. Use what you learned from both sides of the peer review to write a new, shorter skill for SVM model fitting.

Write the skill by hand. Create a `SKILL.md` with a short name and description in YAML frontmatter. Write the entire file yourself, without asking AI to draft or edit it. The instruction body after the closing `---` of the frontmatter must be **no more than 50 lines**, including blank lines. Keep the skill independent of any particular dataset, variable names, or observed results, and load it only when you explicitly invoke it.

Your skill should direct the agent to:

1. judge whether the data are suitable for an SVM analysis;
2. choose which kernels to consider and explain why;
3. identify tuning parameters and a method for tuning them;
4. choose an evaluation method that supports selecting a final model;
5. require a compact figure and table that justify the choices;
6. interpret the fit and report the key numerical results; and
7. prioritize the evidence needed to justify the final model in **no more than one printed page at 11-point font**.

Apply the skill. Explicitly invoke your skill to analyze `saheart.csv`, using the same response, nine predictors, and `famhist` coding as in Question 1. Provide these data details when invoking the skill, not in the skill itself. Let the skill guide the analysis, then check the resulting report and calculations.

Submit the results. Include the one-page, 11-point analysis report in your Homework 09 submission and append the complete contents of your `SKILL.md` afterward. The appended skill is outside the one-page limit. **Invocation:** `/svm-report` Data: `saheart.csv`; response `chd`; predictors: the other nine columns; `famhist` coded 1 = Absent, 2 = Present; seed 432.

Setup. n = 462, 9 predictors, CHD counts 302 / 160 (0 / 1). Stratified split: 346 training, 116 test; 5 stratified CV folds on training data; seed 432. Predictors were standardized with fitting-fold data only. No missing values were handled specially; classes are moderately imbalanced, so I report the confusion matrix, not just error.

Candidates (19). Linear ($C \in \{0.01, 0.1, 1, 10\}$); polynomial degree 2 and 3 ($C \in \{0.1, 1, 10\}$); radial ($\gamma \in \{0.01, 0.05, 0.2\}$, $C \in \{0.1, 1, 10\}$). Fixed: `coef0 = 1`, polynomial `gamma = 1/p`. Linear is the baseline; polynomial and radial test for nonlinearity.

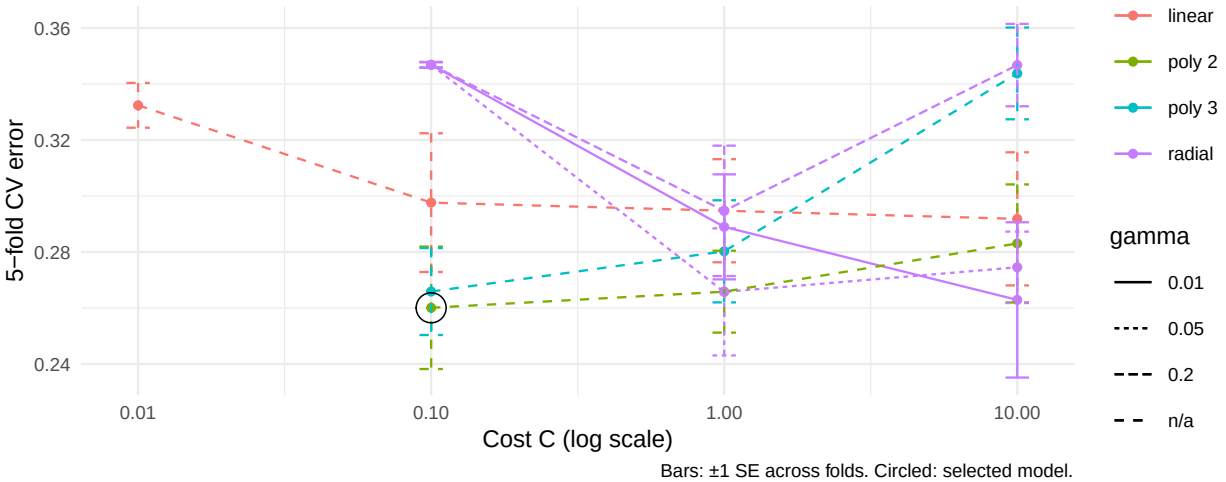

Table 1: Best setting per kernel

Kernel	Degree	Gamma	C	CV error	SE	Selected
linear	NA	NA	10.0	0.292	0.024	

Kernel	Degree	Gamma	C	CV error	SE	Selected
poly 2	2	NA	0.1	0.260	0.022	*
poly 3	3	NA	0.1	0.266	0.016	
radial	NA	0.01	10.0	0.263	0.028	

Results. Selected: poly 2 kernel, $C = 0.1$, CV error 0.26 (SE 0.022), 228 support vectors. Best linear CV error was 0.292 (SE 0.024). Test error: 0.276 (95% CI 0.197–0.367). Test confusion matrix (rows truth, columns prediction): TN = 66, FP = 10, FN = 22, TP = 18.

Conclusion. The selected poly 2 model differs from the linear baseline by 0.032 (1.34 linear-model SEs), so evidence for nonlinearity is moderate.

Limitations. One test split (wide interval at this n), coarse grids, CV SEs from only 5 folds, and nonlinear SVMs give no coefficients to interpret.

Appendix: SKILL.md

```
---
name: svm-report
description: SVM model-selection analysis with a one-page report. Use ONLY when the user explicitly invokes
disable-model-invocation: true
---

# SVM Analysis Report

Fit and justify an SVM classifier using the data, response, predictors, coding, and seed given in the reques

## 1. Suitability
- Check response type, class balance, n vs p, missing values, predictor types and scales, outliers.
- Encode categorical predictors as instructed; standardize using fitting data only.
- State any concern (imbalance, small n, p near n) and how it affects the analysis.

## 2. Kernels
- Consider linear (baseline), low-degree polynomial, and radial (flexible default). Give a data-based reason
- Cap candidates at about 20 total. State all fixed settings (e.g., offset, polynomial scaling).

## 3. Tuning
- Tune cost C on a log grid for every kernel, degree for polynomial, gamma on a log grid for radial.
- Use a coarse grid; extend it if the best value lies on its edge.
- Tune by cross-validation, never by training error.

## 4. Evaluation
- Set and report the seed. Hold out a stratified test set (~25%) before exploring; do not touch it until the
- On the training data, build stratified K-fold splits (K=5 or 10) once; reuse them for every candidate. Fit
- Select the lowest mean CV error. If a simpler candidate (linear, lower degree, smaller gamma or C) is with
- Refit on all training data, evaluate once on the test set; report error with a 95% binomial interval. Neve

## 5. Figure and table (required)
- Figure: CV error with SE bars versus tuning parameter, one line per kernel, selected model marked.
- Table: best setting per kernel with CV error and SE, selected model marked.

## 6. Interpretation
- Report selected kernel and settings, CV error and SE, test error with interval, confusion matrix, and supp
- Compare the selected model with the linear baseline in units of SE; say whether nonlinearity is supported
- Note that nonlinear SVMs give no coefficients; do not over-interpret.

## 7. One-page report
- At most one printed page at 11-point font. Verify it; shrink figure and table, never the font.
- Order: setup, candidates and grids, figure, table, results, conclusion, limitations.
- Prioritize evidence that justifies the final choice. Omit code and raw output. Round to three decimals.
```